

chapter nine

HYPOTHESIS TESTS ABOUT THE MEAN AND PROPORTION

*It is a capital mistake to theorize
before one has data.*

– Arthur Conan Doyle

INTRODUCTION

In the last three chapters we worked with continuous probability distributions. The z score was first presented in chapter 6. We used the standard normal variable z and the related t distribution for small samples to answer probability questions related to the distributions of x , $\bar{x}$ and $\hat{p}$. In chapter 8 we used the z and t scores to extend our work to confidence intervals and estimating the sample size to obtain a degree of accuracy in the interval estimate of the population mean and proportion.

The **confidence interval** was an extension of the work we did in chapter 3, where we found what are called **point estimates** of the population parameter. For example $\bar{x}$ was a point estimate of the population mean. We used both $\bar{x}$ and the z score to find an interval that contained μ . More importantly, we were able to state a level of confidence that the interval we created contained μ a certain percent of the ~~time~~. **value**

This chapter extends our work to what is described as **hypothesis testing**. The idea is simply an extension of what we did in the last two chapters with the z and t score. For example, we used the z score to find the probability that the sample mean was greater than some value. In each problem we were given the value of μ and σ to use in our calculation of z . We then used the z table to find the probability.

In this chapter we are given μ or p as a hypothesized value someone believes to be true. We start by first finding the value of the z score or value of t related to the sample statistic $\bar{x}$ or $\hat{p}$ and the hypothesized value of the population parameter. We then find the probability that the **test statistic** is larger than or less than a **critical value** found in the z table or given by a calculator.

If the probability we find is less than 5% (which is called the level of significance, α), we decide to reject the **null hypothesis**. The value of α states the percent of the time that we are wrong if we reject a null hypothesis that is true. There are several new terms defined that relate to hypothesis testing. It is important to take the time to learn what the terms mean.

An important key to this chapter, and the chapters that follow, is the general structure of hypothesis testing. This process will be used in all of the remaining chapters and involves five parts:

1. State the **null and alternate hypotheses**, H_o and H_a .
2. Select the **distribution** to use.
3. Determine the **rejection and nonrejection regions** using a table.
4. Calculate the value of the **test statistic**.
5. Make a decision to either **reject H_o** or **fail to reject H_o** .

CHAPTER 9 HOT SPOTS

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If you find other HOT SPOTS, write them down and use them as a focus of your discussions in the study group. Or you can use the HOT SPOTS as the topic for a help session with your professor.



Hot Spot #1 – Deciding On The Null Hypothesis, H_0 .

It is fairly easy to decide on the null hypothesis when doing our own research. All we do is let the alternate hypothesis equal what we want to prove. We often refer to the alternate hypothesis as the research hypothesis. The null hypothesis then becomes the complement of the alternate hypothesis or research hypothesis. For example, if we want to prove that the population mean of highway speeds is greater than 55, we write

$$H_a: \mu > 55.$$

The null hypothesis then becomes the complement of > 55 .

$$H_0: \mu \leq 55.$$

We usually write the null hypothesis using the = sign as follows: (Hot Spot #4 discusses why we use the equal sign).

$$H_0: \mu = 55.$$

Technically, a null hypothesis can be written with any one of the following three symbols:

$$\leq \text{ or } = \text{ or } \geq$$

Rather than making it difficult to determine which form to use in the null hypothesis, as a rule of thumb, we just remember that **the null hypothesis has the equal sign in it.**

The alternate hypothesis (called the research hypothesis) then has one of the following symbols in it:

$$< \text{ or } \neq \text{ or } >$$

(Hot Spot #3 discusses how to decide on the choice of symbols for the alternate hypothesis). The choice of the alternate hypothesis really is just a matter of deciding whether the critical region is in one or two tails of the standard normal curve.

Remember: The null hypothesis has the equal sign in it.

Hot Spot #2– Deciding On The Test Statistic

In this chapter, we use the test statistic z in our test of hypothesis. The sample statistic we use with the z test can be in one of three forms. We can use the sample mean ($\bar{x}$), the sample proportion ($\hat{p}$), or the number of successes (x). The choice of the test statistic is made as we read the problem. We are looking for two words, mean and proportion. When a problem involves the proportion we often have to interpret the meaning from the way the problem is written. One of the keys is to look for a percent that is given; the value given will usually be the value of the population proportion used in H_0 .

When we read that we are trying to show something is significant about the population mean, we use

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

If we read that we are trying to show something significant about the population proportion, we use

$$z = \frac{x - np}{\sqrt{npq}} \quad \text{or} \quad z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

The second of these two formulas for z has the advantage of fitting the structure we used earlier:

$z = \frac{\square - \mu_{\square}}{\sigma_{\square}}$	where the $\square$ can be	1. x 2. $\bar{x}$ 3. $\hat{p}$
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We use the z test statistic to make inferences about the population parameter. As we go into the following chapters we will continue to use the z value to test hypotheses about the population parameter.



Mann
Sections 9.2,
9.4, And 9.5

Expanded
Template For z
Test Statistic



Hot Spot #3 – Deciding On A One-Tailed Versus A Two-Tailed Test

The research hypothesis is what we want to prove. We write the research hypothesis as what we call the alternate hypothesis, H_a . We use one of the following three symbols of inequality in H_a :

$<$ or $\neq$ or $>$

We look for key words in the problems we do to decide on the choice of the inequality symbol to use with H_a . The following table shows many of the words that are used with each of the inequality symbols. In each situation the statement in the problem would involve showing that the population parameter was significantly _____, where the blank would be one of the words taken from the table below:

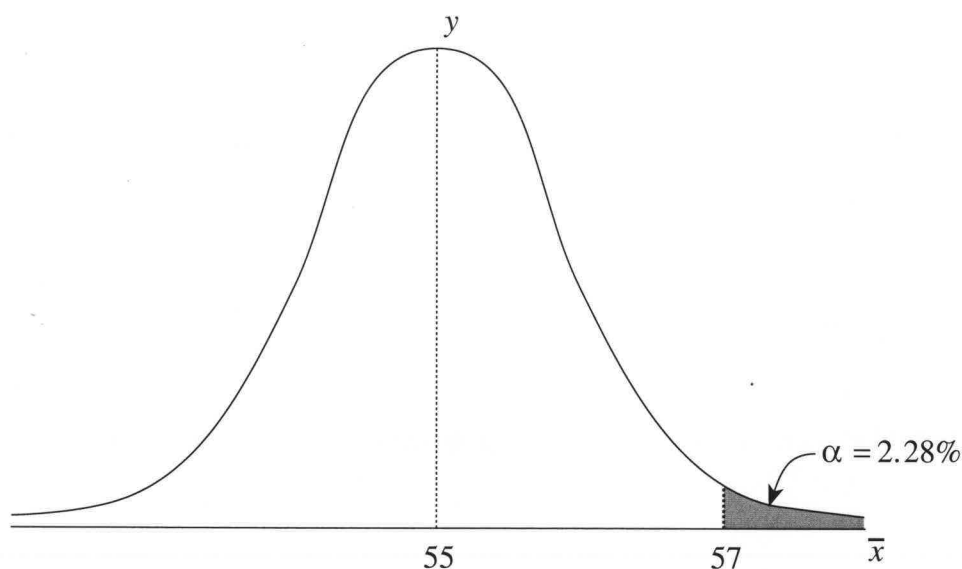
Table of Words and Symbols Used in H_a		
$<$	$\neq$	$>$
Less Than	Different	More Than
Decreased	Changed	Increased
Reduced		Greater
Lowered		Higher
Worse		Better

Table 9-1:
Words And
Symbols Used In
The Alternative
Hypothesis

One of the important points to remember is that the direction of the inequality symbol tells us the location of the critical region. The inequality symbol is like the directional signal in a car. When the arrow points left $<$, we have the critical region on the left side of the curve. When the arrow points right $>$, we have the critical region on the right side of the curve. When the symbol is not equal $\neq$, it is like having the emergency flasher system on where both tail lights blink. We then have the critical region in both tails of the curve.

Hot Spot #4 – Why We Use The = In A One-Tailed Test

It is easiest to look at this question through an example using highway speeds. In a hypothesis test of $H_0: \mu \leq 55$ and $H_a: \mu > 55$ with $\sigma = 5$, a sample of $n = 25$ gave a sample mean of 57. We are interested in the probability that $\bar{x}$ is greater than 57 given that $\mu \geq 55$. We can start by looking at the normal curve with $\mu = 55$. Using the z formula for $\bar{x} = 57$, we get $z = 2$. The area to the right of 57 is $P(z > 2) = 0.0228$, and represents 2.28% of the area under the curve. The probability equal to the area under the tail of the curve is called the significance level of the hypothesis test. We use the symbol α for the size of the significance level.



Next, we will look at the area under the normal curve for a value of $\mu < 55$, that is $H_0: \mu = 53$. Using the z formula for $\bar{x} = 57$, we get $z = 4$. Using the HP-21S, we find that the area to the right of 57 is

$P(z > 4) = 0.0000317$ and represents 0.00317% of the area under the curve. This time we say that $\alpha = 0.0032\%$. The table in Table 7-2 below shows the value of α for different values of μ using the HP-21S.



Mann
Section 9.1.4

Figure 9-1:
Significance
Level Of The
Hypothesis Test

Table 9-2:
Significance
Levels For
Different
Population Mean
Values

Values of α for Different Population Mean Values	
μ	α
55	2.28%
54	0.13%
53	0.0032%
52	0.000029%
51	0.00000010%

When we compare values in the table, we see that for values of $\mu < 55$, the area under the tail of the curve is much less than the area for $\mu = 55$. In this sense, the value of μ that is given by the equal sign in the null hypothesis, H_0 , gives us the worst case scenario for the α value; that is, the largest value of α . Thus, it is common practice to simply write the null hypothesis with the equal sign. In the above example we would write

$$H_0: \mu = 55 \text{ and } H_a: \mu > 55.$$

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Mann
Section 9.1.3

Hot Spot #5 – Choosing A Value Of α

It is important that we recall that the value of alpha (α) is the probability of rejecting a true null hypothesis. When we reject H_0 , we report our findings to others in the form of a research paper or news release. If the null hypothesis is true and we make the mistake of rejecting H_0 and reporting our results to the world, we will be very embarrassed to say the least. Most of us are very worried about making a fool of ourselves in public. So we try to keep the value of α small. We

usually choose a value of 1% or 5% for α . There are times that we may select a value of 10% when we are in the early stage of a research program. It is often difficult to find significant results, and we may be willing to accept a large value of α .

There are other considerations that may affect the value of α . For example, if we plan to publish a research report in a journal, it may be a common and accepted practice that all research be conducted using a value of 5%. In other situations, we may be doing a research project and a committee or advisor may decide on the level of α that should be used.

Another point is involved when we choose the value of α . We will illustrate the point using a right-tailed hypothesis test. The smaller the value of α we choose, the larger the critical value. For example, in a one-tailed test using the z test statistic, the critical value is 1.645 for $\alpha = 5\%$, but it is 2.326 when $\alpha = 1\%$. In order to reject H_0 at 1%, we need a larger value of z . Since

$$z = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}}$$

we will need a larger difference between $\bar{x}$ and μ . In research it is usually very difficult to find large differences between the sample mean and its hypothesized value. Thus, it makes more sense to use the smaller critical value of 1.645 to make it easier to reject H_0 .

It helps to keep in mind the inverse relationship between α and the critical value:

α gets smaller as the critical value gets larger.

In summary, if we must choose the value of α on our own, we use 5%.

Hot Spot # 6 – The Relationship Between α And The Reported P Value.

There is a common practice these days to actually report the significance level at the value given by the test statistic. The following example illustrates the practice:

In a hypothesis test of $H_0: \mu = 55$ and $H_a: \mu > 55$ with $\sigma = 5$, a sample of $n = 25$ gave a sample mean of 57. Test for a significant increase at $\alpha = 5\%$.

The value of the test statistic z is

$$z = \frac{57 - 55}{\frac{5}{\sqrt{25}}} = 2$$

We then find the probability that z is greater than 2:

$$P(z > 2) = 0.0228$$

We say that the reported P value of α is 2.28%.

We now have several choices that we can make:

1. Since the critical value is 1.645, we can report our research findings significant at the predetermined value of $\alpha = 5\%$.
2. We can simply report our research finding at the reported P value of 2.28%.
3. We can combine the two approaches by arguing that the P value of 2.28% is less than $\alpha = 5\%$; hence, we reject H_0 .

There are two disadvantages to the first choice. First, we are not able to pass on the additional information that the reported P value provides, namely how significant our result is. Secondly, there are times when a

reported P value would allow us to report research findings that would otherwise go unreported. For example, if the sample mean in the above hypothesis test was equal to 56.5, the value of the test statistic would have been equal to 1.5; we would fail to reject at $\alpha = 5\%$. However, if we decide to use the reported P value, we find that the probability that z is greater than 1.5 is 0.0668. We could report a significant increase at 6.68% and leave it to the reader to determine if he or she wishes to accept the risk of the type I error.

In the second choice, we seem to have the advantage of not having to worry about a predetermined significance level. However, there is another consideration that we need to address. The value of α (the probability of rejecting a true null hypothesis) is related to the value of β (the probability of failing to reject a false null hypothesis). The two values are inversely related.

α goes down in value as β goes up in value.

Generally speaking, we do not worry too much about the value of β when we fail to reject the null hypothesis.

The type I error associated with α is considered the more serious error. However, β is related to the power of the test. The power of the hypothesis test is the probability of rejecting a false null hypothesis.

$$\text{Power} = 1 - \beta$$

As β gets larger, the power gets smaller. It should be clear that very small values of α ultimately affect the power. As α gets smaller, β gets larger, and power gets smaller.

It is not appropriate to provide quantitative comparisons of the three values at this point in the course. But it is important to understand that one should not allow α to be reported at very low values without considering the possible effect on the power of the test.

In the third choice we are able to combine the best of what is

involved in the first and second choices. When we use the third choice, it is common practice to first state the significance level of the hypothesis test at $\alpha = 5\%$. We then find the reported P value and compare it to α . If $P < \alpha$ then we reject H_0 . The reported P value then has the advantage of showing how significant the result is.

One of the more subtle advantages of using the P value with a stated level of significance is that we do not need to find critical values. If we are using $\alpha = 5\%$, then any value of $P < 5\%$ will allow us to reject H_0 . Note that in the traditional approach to hypothesis testing, we reject when $|z|$ is large, but using the P value we reject H_0 when P is small.

Hot Spot #7– Reporting P Values For The t Distribution

Reporting P values for a t distribution can be a problem when we do not use a calculator like the HP-21S. If we use the calculator, we can easily find the reported P value. All we have to do is enter the calculated value of t and the value for the degrees of freedom and press one key to give us the reported P value. However, P values can be a problem without the calculator.

The reason for the problem is that the table we use for critical values of t lists all the values of t for each degree of freedom under 30 for each value of α . The following table shows a typical row of values in a t table for $df = 9$:

df	$t(0.005)$	$t(0.01)$	$t(0.025)$	$t(0.05)$	$t(0.10)$
9	3.250	2.821	2.262	1.833	1.383

The problem with reporting a P value for a calculated t test statistic is shown in the following example where a hypothesis test was used to find a significant reduction in cholesterol levels for a sample of

10 patients who reduced the amount of fat in their diets. The information used in the hypothesis test was as follows:

$$H_0: \mu = 200 \quad \text{and} \quad H_a: \mu < 200$$

$$\bar{x} = 192 \quad s = 11.35 \quad n = 10$$

$$t = \frac{192 - 200}{\frac{11.35}{\sqrt{10}}} = -2.229$$

Since the alternate hypothesis points to the left side of the curve, we need to think of the values in the above table as representing negative critical values on the left side of the curve. We then compare the calculated (observed) value of 2.229 to the entries in the table. Our observed value is between the critical values for α at $2\frac{1}{2}\%$ and 5% .

$$-2.262 < -2.229 < -1.833$$

We report the significance level as a value of less than 5% but more than $2\frac{1}{2}\%$.

$$2\frac{1}{2}\% < \alpha < 5\%$$

Using the HP-21S, we find that the actual P value is 0.0264.

Hot Spot #7 Sample Problem and Solution: A hypothesis test was done using the information that follows. Find the reported P value.

$$H_0: \mu = 1,200 \quad \text{and} \quad H_a: \mu > 1,200$$

$$\bar{x} = 1,214.75 \quad s = 23.14 \quad n = 20$$

$$t = \frac{1,214.75 - 1,200}{\frac{23.14}{\sqrt{20}}} = 2.851$$

Solution: Step 1. We calculate $df = 20 - 1 = 19$.

Step 2. We refer to the t table at 19 degrees of freedom and find the calculated value of t lies between the critical values of 2.539 at 1% and 2.861 at 0.5%.

Step 3. We report the significance level as less than 1% and greater than 0.5% and write $0.5\% < \alpha < 1\%$.

Sample Problem and Answer: A hypothesis test of $H_0: \mu = 55$ and $H_a: \mu \neq 55$ with $n = 10$ resulted in a value of $t = 1.842$. What is the reported P value?

Answer: Using $df = 9$, we find the calculated t value between 1.833 at 5% and 2.262 at $2\frac{1}{2}\%$. Since the hypothesis test was two-tailed we say that $\frac{\alpha}{2}$ is between $2\frac{1}{2}\%$ and 5%. So the hypothesis test was significant at a value of α between 5% and 10%.

Sample Problem and Answer: A hypothesis test of $H_0: \mu = 220$ and $H_a: \mu > 220$ with $n = 30$ resulted in a value of $t = 2.284$. What is the reported P value?

Answer: Using $df = 29$, we find the calculated t value between 2.045 at $2\frac{1}{2}\%$ and 2.462 at 1%, we say that the hypothesis test was significant at a value of α between 1% and $2\frac{1}{2}\%$.

Hot Spot #8 – The Difference Between Reject And Fail To Reject

It is fairly easy to figure out what it means to reject the null hypothesis. Failing to reject is just the flip side of the coin. The statement "Fail To Reject" is **not**, however, the same as accepting the null hypothesis. When we reject H_0 , we do so at the significance level called alpha, α . The significance level states the percentage of the time we are making a type I error by rejecting H_0 , when it is in fact true.

When we make the statement that we "Fail To Reject H_0 ", we are not obligated to give an error analysis. We have not in effect made any decision about the null hypothesis. If we, however, decided to accept H_0 , we would need to indicate the percentage of the time we are making an error. The error in accepting a false H_0 is called a type II error, and the probability of making the error is called beta, β . Even though we agree on a significance level (α) at the beginning of a hypothesis test, we seldom know the value of β . Therefore, when the test statistic does not fall in the critical region, the best we can do is say that we fail to reject H_0 .

Hot Spot #9 – The Relationship Between α , β , And n

The relationship between α and β is essentially that as one goes up in value, the other goes down in value. We often say that α and β are inversely related. However, the relationship is **not** like inverse variation; the product does **not** equal a constant. Furthermore, the sum of α and β is not 1. It is better to think of the two values as the opposite ends of a "teeter-totter".



Mann
Section 9.1.2, 9.2



Mann
Section 9.1.3

Figure 9-2:
Illustration Of
The Relationship
Between alpha
and beta

As we push down on one side of the “teeter-totter”, that is make α larger (heavier), the β value will become lighter (smaller). It is this inverse relationship that causes problems when we make the value of α extremely small. The value of β will then become quite large. This would be a bad situation because the power of the test of hypothesis is equal to $1 - \beta$. The power of the hypothesis test is the probability of rejecting a false null hypothesis; which is what we want to do.

One of our real concerns is then how to lower both α and β . The solution to our problem is quite easy in one sense of the word. All we have to do is increase the sample size we use in getting the sample mean or proportion. As easy as it sounds, using a larger sample size creates a number of related problems. Large sample sizes require more time, money, and energy. And at times it simply is neither practical nor possible to use a large sample.

The following problem using the binomial distribution shows how the values of α , β , and n are related. We will use $n = 10$ and $p = 0.5$ and set up the hypotheses as follows:

$$H_0: p = 0.5 \text{ and } H_a: p < 0.5$$

We will use a decision rule to reject H_0 if $x \leq 3$. The probability of the type I error is determined by finding $P(x \leq 3)$ when $p = 0.5$ and $n = 10$.

$$\alpha = P(x \leq 3) = 0.1719$$

We need a specific value of $p < 0.5$ to find the value of β . We will choose $p = 0.2$ to illustrate the process. We need to find $P(x > 3)$ when $p = 0.2$ and $n = 10$ to find the probability of the type II error.

$$\beta = P(x > 3) = 1 - P(x \leq 3) = 1 - 0.8791 = 0.1209$$

Although β is at an acceptable level (Power = $1 - \beta = 88\%$), α is too large. We let $n = 5 \times 10 = 50$ and use a decision rule to reject H_0 if $x \leq 15$; the ratio of 15 to 50 is the same as the ratio of 3 to 10.

The probability of a type I error is $P(x \leq 15)$ when $p = 0.5$ and $n = 50$.

$$\alpha = P(x \leq 15) = 0.0033$$

The value of β is found by taking $P(x > 15)$ when $p = 0.2$ and $n = 50$.

$$\beta = P(x > 15) = 1 - P(x \leq 15) = 1 - 0.9692 = 0.0308$$

We now find that we have lowered both the α and β values as a result of increasing the value of n from 10 to 50. Increasing n to even higher values will lower the values of α and β even more.

Now let us see what happens if we leave n at 10 but lower the α value. We will use the decision rule to reject H_0 if $x \leq 2$. We get the probability of the type I error by finding $P(x \leq 2)$ when $p = 0.5$ and $n = 10$.

$$\alpha = P(x \leq 2) = 0.0547$$

We find the probability of the type II error by finding the $P(x > 2)$ when $p = 0.2$ and $n = 10$.

$$\beta = P(x > 2) = 1 - P(x \leq 2) = 1 - 0.6778 = 0.3222$$

Although we lowered the value of α from 0.1719 to 0.0547, we increase the value of β from 0.1209 to 0.3222. In contrast, when we used a larger value of n we were able to lower α from 0.1719 to 0.0033 and lower β from 0.1209 to 0.0308.

The error analysis we just went through was reasonably easy to do for two reasons. The use of the HP-21S made it easy to answer the probability questions and we used a specific value of $p < 0.5$ from the alternate hypothesis. Generally speaking, we do not know a specific value for the alternate hypothesis. As a result, we seldom go through an error analysis. We simply assume that if α is kept at 5% and $n \geq 30$, the value of β will be small enough for us to have a powerful test.

Hot Spot #10 – Difference Between t Test Statistic And z Test Statistic

The formula we use for the t test statistic looks like the z test statistic we used when we substituted the sample standard deviation s for σ , the population standard deviation. This was not a problem as long as the sample size was larger than 30. When the sample size is less than or equal to 30, the sample standard deviation tends to underestimate σ . This can happen because small samples do not often include extreme values from the tails of the distribution. The effect on z , the standard score, is to make the value larger than it should be. When this happens in a hypothesis test, we increase our chance of rejecting a true null hypothesis and making a type I error.

The way we avoid making unnecessary type I errors when we do hypothesis tests with small samples is to adjust the critical value for the sample size. For example, the critical value of z for a test of hypothesis at 5% is 1.645 for a one-tailed hypothesis test on the right side of the curve. But if the sample size is reduced to 10, the critical value is increased to 1.833. As the sample sizes become smaller we continue increasing the size of the critical value. For example, the critical value becomes 2.132 when $n = 5$ and 2.920 when $n = 3$.

We get the new critical values using what is called the student's t distribution. The shape of the curve for the t distribution is bell-shaped looking like the now familiar standard normal curve used with the z table. But the curve for the t distribution is flatter at the center and the tails of the curve seem to float above the horizontal axis when compared to the standard normal curve. As the sample sizes get smaller, the curve becomes even more flattened. We say that the shape of the curve depends on what is called degrees of freedom. The degrees of freedom equal one less than the sample size. We use the letters df to represent degrees of freedom and write the following expression:

$$df = n - 1$$

The graph below shows a general comparison between the standard normal curve for z and student's t distribution curve used for t :

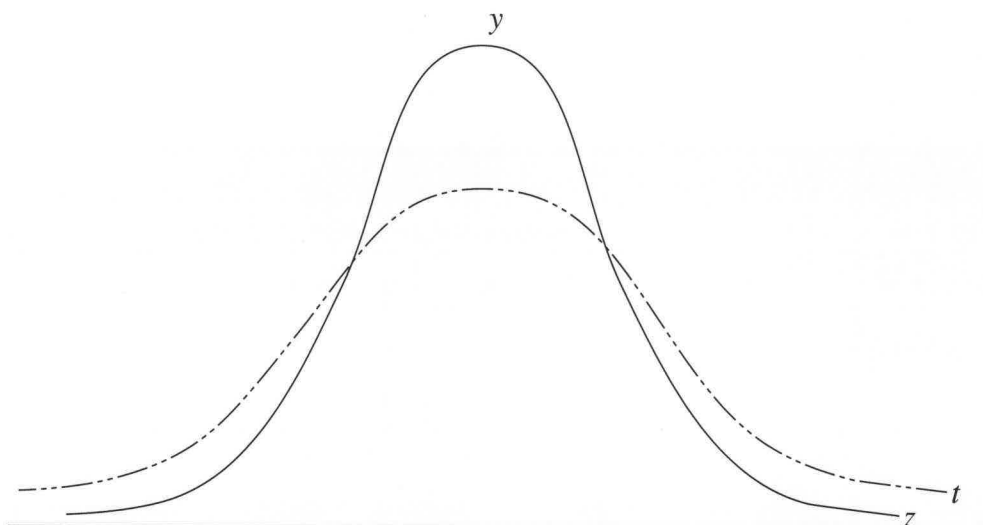


Figure 9-3:
Comparison Of
 z Distribution To
 t Distribution

Actually, we get a different t curve for each value of df . As a result, it is more difficult to list the critical values for the t distribution. One way this could be done is to list a new table for each value of df . But this would require at least 30 tables just to do hypothesis tests. It is now common practice to simply list the critical values for the more common significance levels. The critical value could then be found by finding the appropriate degrees of freedom in a vertical column and then moving across to the required level of α . This allows us to get by with one table of values as opposed to carrying around a small book of tables.

There is a down side to using a single t table when it comes to reporting P values for a hypothesis test. This is discussed in **Hot Spot #7**, but an easy solution to the problem of using the t table is to use a calculator like the HP-21S. It makes it easy to both get a critical value for any degrees of freedom and report the P value for a hypothesis test. The following discussion assumes that we will be using the traditional t table to find critical values.

In the traditional t table, the critical values are listed for each of the commonly accepted significance levels for each value of df below 30.

The table below shows the critical values of t for a 5% one-tailed hypothesis test as they would appear:

Critical Values of t for a 5% One-Tailed Hypothesis Test			
df	$t(0.05)$	df	$t(0.05)$
1	6.314	16	1.746
2	2.920	17	1.740
3	2.353	18	1.734
4	2.132	19	1.729
5	2.015	20	1.725
6	1.943	21	1.721
7	1.895	22	1.717
8	1.860	23	1.714
9	1.833	24	1.711
10	1.812	25	1.708
11	1.796	26	1.706
12	1.782	27	1.703
13	1.771	28	1.701
14	1.761	29	1.699
15	1.753		

Table 9-3:
Critical Values
Of t For A 5%
One-Tailed
Hypothesis Test

It is common practice to simply use the value from the z table for values of df greater than 29. Therefore, we would use $t = 1.645$ for any $df > 29$. It should again be noted that calculators like the HP-21S now allow us to find the critical value for any degree of freedom and any value of α .

If we look carefully at the two curves shown below in Figure 9-4, we can see why the critical value is larger for the t test statistic. The two curves show the critical value for a 5% one-tailed test of hypothesis. The curve used for t is at 9 degrees of freedom. We can see that we must go

out further in the tail of the t curve to get 5% of the area in the tail of the curve.

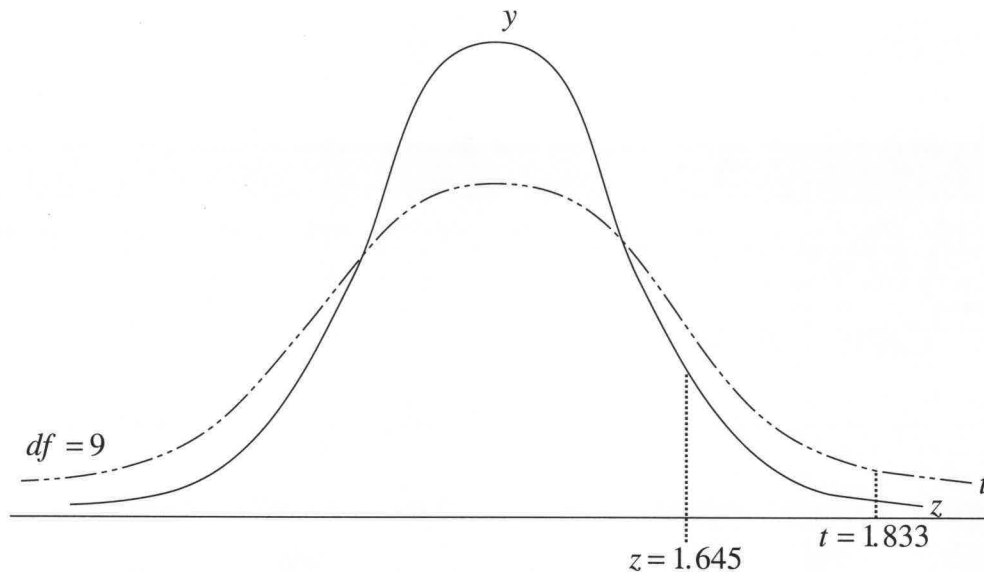


Figure 9-4:
5% Tails Of
 z Distribution
Versus
 t Distribution

The test statistic for the t distribution looks exactly like the z test statistic when we substitute s for σ .

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

We communicate that we are making the adjustment for a small sample by using the letter t for the standard score. Again, remember that we have to make the adjustment because:

1. We do not know the value of the population standard deviation.
2. The sample size is under 30.

If we know σ , the sample size is no longer a problem. When we know σ , we can use the standard normal curve and find the critical value using the z table, regardless of the sample size.

CHAPTER 9 DISCUSSION QUESTIONS

These questions may be used in your study group or simply as topics for individual reflection. Whichever you do, take time to explain verbally each topic to insure your own understanding. Since these questions are intended as topics for discussion, answers to these questions are not provided. If you find that you are not comfortable with either your answers or that your group has difficulty with the topic, take time to meet with your professor to get help.

1. What is the difference between a t and z test statistic?
2. How do you decide which side of the curve the critical value is on?
3. When is a sample considered small?
4. What is the difference between the null and the alternate hypothesis, and how do you choose the null hypothesis?
5. When do you reject the null hypothesis?
6. What is the difference between a type I and type II error?
7. Why is it important to graph the curve and locate the critical value(s) on the standard normal curve when doing a test of hypothesis?
8. What is the difference between a one-tailed and two-tailed test of hypothesis?
9. When do you use the t distribution?
10. What is the difference between α and the reported P value?